Marwan Mohamed Elfouly

+20 112 522 3161 | marwan.mohamedelfouly@gmail.com | linkedin.com/in/marwan-mohamed-atef-elfouly
github.com/Marwan-Mohamed14 | marwan-mohamed14.github.io/portfolio

EDUCATION

Misr International University Cairo, Egypt
Master of Science (M.Sc.), Computer Science *September 2026 – Present*
– Continuing advanced coursework in computer science

Misr International University Cairo, Egypt
Bachelor of Science (B.Sc.), Computer Science — Software Engineering Track *May 2026*
– Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Database Systems, Machine Learning, OOP, Computer Networks

EXPERIENCE

Computer Science Teaching Assistant September 2026 – Present
Misr International University (Full-Time) *Cairo, Egypt*
– Teaching undergraduate computer science courses, covering core technical subjects

Java Developer Intern July 2026 – August 2026
ITWorx *Cairo, Egypt*
– Developed backend services in Java and Spring Boot to build internal tools used in production by the company
– Built Retrieval-Augmented Generation (RAG) pipelines with Spring AI, integrating LLM-powered features into internal tools

ERP Developer — Graduate Program July 2026 – September 2026
Health Insights *Cairo, Egypt*
– Building and customizing ERP modules as part of the company's graduate development program to support core business workflows

Cybersecurity Intern July 2025 – September 2025
Siemens *Cairo, Egypt*
– Evaluated OT/ICS security architectures and conducted penetration testing on live PLC hardware using Kali Linux, solving 30+ CTF challenges focused on ICS security

Instructor December 2024 – June 2025
Waleed Academy *Cairo, Egypt*
– Designed CS curriculum and produced 75+ bilingual (Arabic/English) video lessons on algorithms, OOP, and Linux

Odoo Developer Intern August 2024 – September 2024
Siemens *Cairo, Egypt*

PROJECTS

AI Contract Validation System — Graduation Project
React, Spring Boot, OCL, MySQL | github.com/Marwan-Mohamed14/softwareguard-graduation-project
– Architected Spring Boot microservices and a React frontend for contract validation, with document ingestion, OCL rule evaluation, and JWT-based auth modules
– Built RESTful APIs with role-based authorization, integrating an OCL rule engine with an AI model to detect legal issues in contract clauses

AI Fitness Trainer
Flutter, Dart, Groq AI (LLaMA 3.3 70B), Supabase | github.com/Marwan-Mohamed14/AI_Fittness_Trainer
– Built a cross-platform app generating personalized workout and diet plans based on user goals, age, and preferences
– Implemented progress tracking, gym discovery, water and meal reminders, and a real-time social community feed

Football Analytics Dashboard & Match Prediction Model
Power BI, Python, Scikit-learn | github.com/Marwan-Mohamed14/Ai_model_matches_prediction
– Built a Power BI dashboard analyzing 58,000+ match records across global clubs, enabling data-driven scouting through 15+ tactical profiling metrics including pace, passing, and defensive output
– Trained Random Forest and Decision Tree models to predict match outcomes and expected goals using rolling 5-match form and betting odds from 8+ bookmakers

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, Dart, C#, HTML/CSS
Frameworks: Spring Boot, Spring AI, .NET, RESTful APIs, Flutter, Node.js, React, Odoo
Databases: MySQL, Firebase, Supabase, MongoDB
AI / ML: RAG Systems, LLM Integration, Machine Learning, Fine-Tuning, Agents, OCL Rule Engine
Tools & Systems: Git, Power BI, Selenium, Linux, Kali Linux

CERTIFICATIONS & LANGUAGES

Flutter Developer Certificate — Digital Egypt Pioneers Initiative (DEPI), MCIT 2025
– Completed a 6-month Flutter & Dart program with a parallel Business English track, building a tour-guide app and a metro navigation app

Languages: Arabic (Native) | English (C1) | German (B1)